

Exercise

CFD Simulation for Energy Systems

FBR Project Task Update

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C. Fluidized Bed

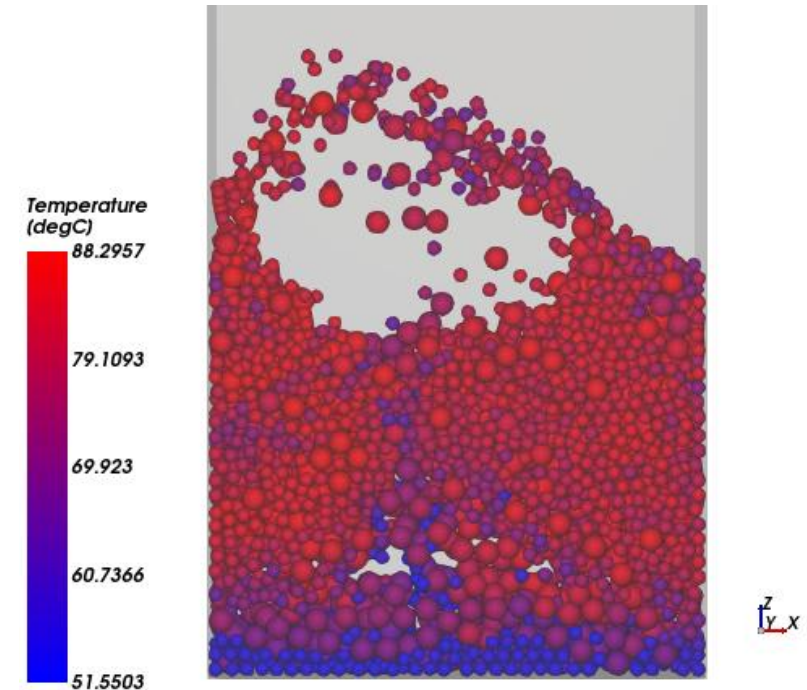
Details:

- Bed dimensions: 0.28 m (W) × 1.0 m (H)
 - Particle diameter: $d_p = 1\text{-}5\text{ mm}$ (adjust for a reasonable total number of particles and simulation time for the given reactor dimensions), $\rho_p = 1500\text{ kg/m}^3$
 - Initial bed height: $H_0 = 0.40\text{ m}$, solid volumetric fraction: $\alpha_p = 0.60$
 - Gas (air): $\rho_g = 1.2\text{ kg/m}^3$, $\mu = 1.8 \times 10^{-5}\text{ Pa}\cdot\text{s}$
 - Reynolds regimes varied via inlet velocity U_0
- How does the flow regime change with Re ?
 - Where do bubbles form and how do they coalesce?
 - How is the solid volume fraction distributed axially?

Minimum convergence criterion of the residuals: 10^{-3}

ANSYS Documentation (Chapter 14: Tutorial - Fluidized Bed):

https://ansyshelp.ansys.com/public/account/secured?returnurl=/Views/Secured/corp/v252/en/dem_tut/dem_tut_14.html



C. Fluidized Bed

Load case as instructed in “Chapter 14: Tutorial - Fluidized Bed” – see link below. Follow the instructions to complete the following tasks:

- a. Set transient solver. Scale geometry to meters. Identify and rename inlet, outlet, and wall zones if necessary.
- b. Enable Eulerian multiphase (gas–solid). Set k- ϵ turbulence for gas phase. Define granular solid phase: $d_p = 3\text{-}5\text{ mm}$, $\rho_p = 1500\text{ kg/m}^3$.
- c. Explain what is minimum fluidization velocity. Estimate minimum fluidization velocity for your case.
- d. Gas inlet: velocity U_0 (varied per regime). Outlet: pressure. Walls: no-slip. Initialize packed bed ($\alpha_p = 0.60$) up to $H_0 = 0.40\text{ m}$.
- e. Run at least 1 case that was assigned to your group (see Excel file on Moodle): Bubbling, Turbulent, or Fast fluidized.
- f. Plot time-averaged solid volume fraction, gas velocity vectors, and particle temperature. Report these at least at a) start b) middle c) end of the simulation. Make sure all phenomena of interest are captured in these plots.
- g. Compute pressure-drop curve; compare with Ergun correlation for packed-bed region.
- h. Briefly explain the difference between 1-way, 2-way, and 4-way coupling. How is particle modeling with ANSYS Rocky different from particle modeling in ANSYS Fluent? Why not using finite volume methods for particle modeling?
- i. Using slightly different parameters than given here is allowed. **However, always report what exact values were used in your simulations.**
- j. Own tests are welcome 😊

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https://ansyshelp.ansys.com/public/account/secured?returnurl=/Views/Secured/corp/v252/en/dem_tut/dem_tut_14.html